

Qatar-2b

R-0669

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_260405 · created 2026-07-27T18:22:21+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461135.8597 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.065
Transit depth (Rp/Rs)^2	0.00012 +/- 0.014 [%]
Orbital Inclination (inc)	88.63 +/- 1.74
Airmass coefficient 1 (a1)	0.0174 +/- 0.0098
Airmass coefficient 2 (a2)	2.95 +/- 1.2
Scatter in the residuals of the lightcurve fit is	53.88 %
Best Comparison Star	None
Optimal Aperture	2.06
Optimal Annulus	7.71
Transit Duration (day)	0.0675 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.065 vs 0.16327 (deviation 2.5σ)
Duration fit vs published	0.0675 d vs 0.0767 d (deviation 2.41σ)
Mid-time O-C	-2.5 min
Depth SNR	0.0
Residual scatter	53.88 %

Light curve

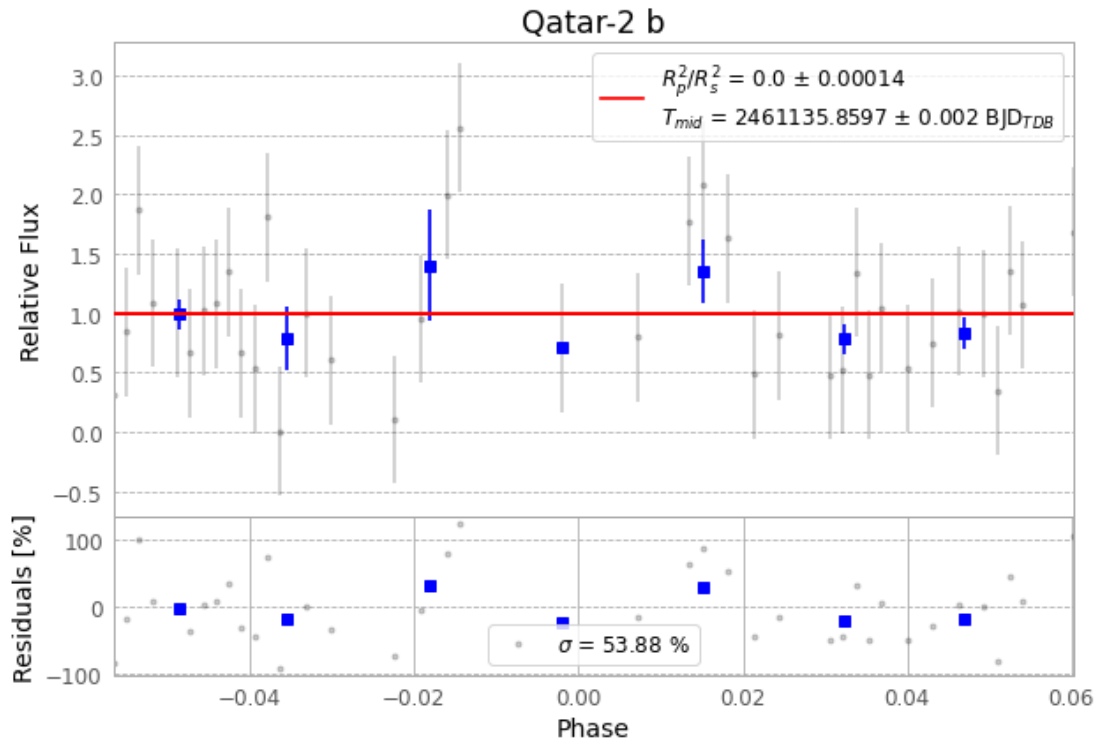


Figure 1 — FinalLightCurve_Qatar-2 b_2026-04-04.png (SHA-256 5f08f4c19cea4536...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 343], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3dd765fac487e867...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f6a95f5

Data provenance

77 FITS frames (51 MB), Qatar-2260405063917.FITS → Qatar-2260405102715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-260405.log (fb5b32f54929...), FinalLightCurve_Qatar-2 b_2026-04-04.png (5f08f4c19cea...), FinalLightCurve_Qatar-2 b_2026-04-04.csv (6b61d203fded...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 18:22Z.